



Structured PhD in Statistics, University of Galway (4 years, full time, fully funded)

Project supervisor: Dr Nicola Fitz-Simon, School of Mathematical and Statistical Sciences, University of Galway.

Project title: Statistical methods to improve causal effect estimation at the small area level with applications in public health policy

Project details: Observational causal inference helps guide public health policy when randomised trials are not possible or ethical. However, standard observational studies often cannot provide reliable results for small subgroups, where treatment effects may differ. Small area estimation (SAE) combines survey data with auxiliary information to improve estimates for small populations. However, research linking causal inference and SAE remains relatively undeveloped.

This project will develop and extend statistical methodology for causal small-area estimation using observational data, motivated by applications such as the WHO STEPS surveys of non-communicable disease risk factors in lower- and middle-income countries. The research will develop statistical methodology, integrating propensity score-based causal inference with established SAE frameworks, including Fay-Herriot and nested error regression models, with emphasis on robust and efficient estimation strategies, including doubly robust approaches.

Key challenges motivating the methodological work include, but are not limited to: handling multiple treatment levels beyond simple treated-untreated comparisons; addressing missing or unmeasured variables; and correcting for measurement error in survey and biological measurements that may bias causal estimates. The project offers scope for methodological development in areas including robust estimation, causal inference, hierarchical modelling, and computational statistics. Overall, the project aims to deliver robust causal SAE methodology for observational data, to strengthen evidence for policy and population-level health decision-making.

Funding details: Stipend of €25,000 per year (tax free) plus additional funding for a laptop. Fees for EU and non-EU students are covered.



The candidate: You should be independent, motivated, and genuinely interested in developing new statistical methods and applying methods using real-world data.

Essential criteria:

Applicants should have, or expect to attain before the project start date, a 2.1 honours degree or higher in statistics, mathematics, or a closely related discipline. Applicants should have excellent R programming skills, or strong programming skills in another language together with a willingness to learn R. Applicants should also have excellent communication skills, and, where relevant, should meet the English language proficiency requirements of the University of Galway.

Desirable criteria:

An MSc in a relevant discipline. Relevant research or practical experience in areas such as hierarchical modelling, causal inference, survey statistics, or Bayesian statistics.

Application Instructions:

Please submit a single pdf file consisting of:

- (1) A letter of motivation, explaining why you are interested in this position, and demonstrating how this position aligns with your experience, skills, and training to date.
- (2) A CV, including the contact details for two academic referees.
- (3) Degree transcripts.

Please submit your application by email to nicola.fitz-simon@universityofgalway.ie, with subject line "PhD SAE/causal".

Application deadline: Applications will be reviewed on a rolling basis, commencing in May 2026, until the position is filled. Only shortlisted applicants will be contacted.

Start date: 1st September 2026, or as soon as possible thereafter.